

IFRS 17 Term Life Valuation Engine

Actuarial Methodology and Validation Report

Prepared by	[Name Placeholder]
Project type	Actuarial Science CV / Portfolio Project
Date	[Date Placeholder]
Technology	Python 3.11+, pandas, NumPy, Pydantic, pytest, openpyxl

This document is prepared as a professional methodology and validation report for an academic actuarial portfolio project. It does not represent a production IFRS 17 reporting system or insurer-approved methodology.

Table of Contents

1. Executive Summary
 2. Project Objective and Scope
 3. Product and Portfolio Description
 4. Data, Assumptions, and Synthetic Data Rationale
 5. System Architecture
 6. Projection Methodology
 7. Cashflow Methodology
 8. Discounting Methodology
 9. Best Estimate Liability Methodology
 10. Risk Adjustment Methodology
 11. Contractual Service Margin Methodology
 12. IFRS 17 Grouping and Onerous Contract Testing
 13. Scenario Analysis
 14. Audit, Governance, and Reproducibility Features
 15. Validation and Testing Framework
 16. Key Strengths of the Project
 17. Limitations and Future Improvements
 18. Conclusion
- Appendix A. Main Configuration Parameters
- Appendix B. Output Files
- Appendix C. Test Coverage Summary
- Appendix D. CV Project Description

1. Executive Summary

The IFRS 17 Term Life Valuation Engine is a modular Python-based actuarial valuation project designed to demonstrate simplified IFRS 17 measurement concepts for term-life insurance contracts. The project includes policy-level projection, mortality and lapse modelling, cashflow generation, discounting, Best Estimate Liability calculation, Risk Adjustment, Contractual Service Margin calculation and roll-forward, onerous contract testing, IFRS 17-style grouping, scenario analysis, audit snapshots, output generation, and automated validation testing.

The project is intended as an academic and CV portfolio project for a 4th-year Actuarial Science student. It is not presented as a production IFRS 17 reporting platform. Production use would require formal methodology approval, data governance, assumption calibration, independent model validation, detailed financial reporting controls, audit workflow, data lineage management, and company-specific IFRS 17 accounting policy decisions.

A notable feature of the project is the inclusion of automated testing. The current test suite includes 17 passing tests covering module imports, projection schema, discount curve integrity, cashflow and BEL smoke checks, present value timing conventions, CSM calculations, grouping consistency, scenario directionality, and audit output determinism.

The use of synthetic data is appropriate for this type of actuarial science project. Real policy-level insurance datasets are typically confidential and unavailable for public academic use because they may contain sensitive demographic, financial, underwriting, claims, and company experience information.

Overall, this project represents a strong actuarial science portfolio project. It demonstrates practical Python modelling ability, actuarial valuation thinking, IFRS 17 awareness, testing discipline, and an understanding of the distinction between educational modelling and production-grade financial reporting.

2. Project Objective and Scope

2.1 Objective

The objective of this project is to build a modular actuarial valuation engine for term-life insurance contracts under a simplified IFRS 17-style measurement framework.

- Policy-level projection logic
- Mortality and lapse assumption application
- Insurance cashflow generation
- Present value cashflow calculation
- Best Estimate Liability calculation
- Simplified Risk Adjustment calculation
- Initial CSM and CSM roll-forward logic
- Onerous contract testing
- IFRS 17-style grouping
- Scenario analysis under shocked assumptions
- Audit and reproducibility outputs
- Automated testing and regression checks

2.2 Scope

Area	Included in Project
Policy-level projection	Yes
Mortality modelling	Yes
Lapse modelling	Yes
Premium cashflows	Yes

Area	Included in Project
Claim cashflows	Yes
Expense cashflows	Yes
Reinsurance cashflows	Yes
Discounting	Yes
BEL calculation	Yes
Risk Adjustment	Simplified educational method
CSM calculation	Simplified educational method
IFRS 17 grouping	IFRS 17-style grouping
Scenario analysis	Yes
Audit metadata	Lightweight audit outputs
Automated tests	Yes
Production IFRS 17 reporting	No

2.3 Scope Disclaimer

This project is not a production IFRS 17 reporting system. It should be interpreted as an educational actuarial modelling project. A production IFRS 17 system would require formal actuarial methodology approval, company-specific IFRS 17 accounting policy, experience-based assumption calibration, source system reconciliation, independent model validation, change control workflow, and financial statement disclosure production.

3. Product and Portfolio Description

The model is designed around a simplified term-life insurance product. A term-life insurance contract typically provides a death benefit if the insured life dies during the policy term. Premiums may be paid periodically, and the insurer is exposed to mortality, lapse, expense, and discount-rate risks.

Feature	Description
Sum assured	Death benefit amount payable on insured death
Premiums	Gross and net premium inflows
Mortality	Probability of death during each projection period
Lapse	Probability of policy termination before maturity
Expenses	Operating expenses associated with policy administration
Reinsurance	Ceding, recovery, and counterparty default components
Projection years	Annual policy projection structure
Discounting	Present value calculation using timing conventions
IFRS 17 grouping	Portfolio, cohort, risk, and profitability classification

The model operates at policy level and then aggregates results for valuation, grouping, scenario, and reporting purposes. A policy-year grid is created so that each policy is projected across multiple future years.

4. Data, Assumptions, and Synthetic Data Rationale

The project uses synthetic policy data and sample actuarial assumptions. The policy data is generated for modelling and demonstration purposes rather than extracted from a real insurer administration system. The model may include a sample mortality table and fallback mortality logic to support educational use where a real company mortality table is unavailable.

Synthetic data is appropriate and professionally reasonable for this type of actuarial science portfolio project. Real policy-level life insurance datasets are usually confidential and not publicly available because they may include personal policyholder information, age and demographic details, underwriting information, policy financial details, premium history, claims history, lapse behaviour, company-specific experience data, reinsurance arrangements, and internal pricing or reserving assumptions.

Because of confidentiality, privacy, and commercial sensitivity, actuarial students rarely have access to real policy-level insurance datasets suitable for public GitHub projects. Synthetic data allows the project to demonstrate actuarial model architecture, projection methodology, cashflow logic, discounting mechanics, IFRS 17-style valuation structure, testing discipline, and reproducible Python modelling.

The important distinction is that synthetic data can demonstrate methodology and coding ability, but it cannot prove that assumptions are calibrated to real insurer experience. In a production setting, assumptions would need to be formally approved, version-controlled, documented, and supported by experience studies or expert judgement.

5. System Architecture

The project is organised into a modular structure. This supports reviewability, maintainability, and professional presentation.

Directory / File	Purpose
config/	Model assumptions and run configuration
data/	Sample mortality table and input data
engine/	Core calculation modules
model/	Pydantic configuration model
tests/	Automated validation and regression tests
Formulas/	Formula and methodology notes
outputs/	Generated outputs, ignored by git
main.py	End-to-end engine runner

Module	Main Responsibility
engine/projection.py	Creates synthetic master data and policy-year projection
engine/curves.py	Builds discount and lapse curves
engine/cashflows.py	Calculates premiums, claims, expenses, reinsurance, and PV cashflows
engine/bel.py	Calculates Best Estimate Liability
engine/ra.py	Calculates simplified Risk Adjustment
engine/csm.py	Calculates initial CSM and CSM roll-forward
engine/grouping.py	Performs IFRS 17-style grouping and onerous classification
engine/scenarios.py	Performs scenario analysis under shocked assumptions
engine/outputs.py	Produces CSV, Excel, audit, and registry outputs
engine/audit.py	Creates assumption snapshots, audit reports, and run metadata

Configuration

- > Discount and lapse curves
- > Synthetic policy master data
- > Policy-year projection
- > Mortality and lapse assumptions
- > Cashflow generation
- > Present value cashflows
- > BEL calculation
- > Risk Adjustment calculation
- > CSM calculation and roll-forward

- > IFRS 17-style grouping
- > Scenario analysis
- > Output files and audit metadata
- > Automated validation tests

6. Projection Methodology

The projection module creates a policy-year grid. For each policy and projection year, the engine calculates current age, issue year, portfolio identifier, in-force status, mortality rate, lapse rate, survival multiplier, opening survival ratio, and sum assured.

The annual survival multiplier is defined as:

$$\text{survival_multiplier}(t) = (1 - q_x(t)) \times (1 - \text{lapse_rate}(t))$$

The survival ratio represents the proportion of policies expected to survive to the opening of a projection year:

$$\text{survival_ratio}(t) = \text{product of survival_multiplier from year 1 to year } t - 1$$

Therefore:

$$\text{survival_ratio}(1) = 1$$

This is an important timing convention. At the start of year 1, the policy is assumed to be fully in force before mortality or lapse decrements are applied. The projection methodology is simplified but actuarially meaningful because expected future cashflows depend on in-force status, probability of death, probability of lapse, policy duration, age, and sum assured.

7. Cashflow Methodology

Cashflow Component	Direction	Description
Gross premium inflow	Inflow	Premium received before deductions
Net premium inflow	Inflow	Premium after relevant adjustments
Death benefit	Outflow	Expected claim payment on death
Surrender benefit	Outflow	Benefit payable on lapse or surrender
Operating expenses	Outflow	Policy administration and operating costs
Reinsurance ceding	Outflow	Premium or cost ceded to reinsurer
Reinsurance recovery	Inflow	Expected recovery from reinsurer
Counterparty default cost	Outflow	Expected cost of reinsurer default
Total inflows	Inflow	Sum of relevant inflow components
Total outflows	Outflow	Sum of relevant outflow components
Net cashflow	Net	Total inflows less total outflows, or equivalent model convention

Premium inflows are modelled as expected cashflows linked to policy in-force status. Death benefits are calculated using mortality assumptions, survival exposure, and sum assured.

$$\text{expected_death_benefit}(t) = \text{sum_assured} \times \text{survival_ratio}(t) \times q_x(t)$$

Operating expenses are treated as expected outflows. Reinsurance is represented through ceding, recoveries, and counterparty default cost. This is a useful educational feature because reinsurance can materially affect fulfilment cashflows and risk exposure.

8. Discounting Methodology

The model discounts future cashflows using an annual discount rate.

$$\text{discount_factor}(t) = 1 / (1 + r)^t$$

$$\text{discount_factor_opening}(t) = 1 / (1 + r)^{(t - 1)}$$

Cashflow Type	Discounting Timing
Premium inflows	Opening discount factor
Reinsurance ceding	Opening discount factor
Claims	Closing discount factor
Expenses	Closing discount factor
Surrender benefits	Closing discount factor
Reinsurance recoveries	Closing discount factor
Counterparty default costs	Closing discount factor

This distinction is actuarially important. Premiums are often assumed to be received at the beginning of the period, while claims and expenses may be assumed to occur later in the period. A production IFRS 17 model would normally require more detailed treatment of current discount curves, locked-in discount curves, illiquidity premium methodology, currency-specific curves, yield curve governance, and OCI versus P&L; accounting policy.

9. Best Estimate Liability Methodology

The model uses a liability-positive Best Estimate Liability convention:

$$\text{BEL} = \text{PV}(\text{outflows}) - \text{PV}(\text{inflows})$$

BEL Value	Interpretation
BEL > 0	Net liability
BEL < 0	Net asset

$$\begin{aligned} \text{PV}(\text{outflows}) &= \text{PV}(\text{claims}) + \text{PV}(\text{expenses}) + \text{PV}(\text{surrender benefits}) \\ &\quad + \text{PV}(\text{reinsurance ceding}) + \text{PV}(\text{counterparty default costs}) \end{aligned}$$

$$\text{PV}(\text{inflows}) = \text{PV}(\text{premiums}) + \text{PV}(\text{reinsurance recoveries})$$

$$\text{BEL} = \text{PV}(\text{outflows}) - \text{PV}(\text{inflows})$$

The BEL calculation includes adjusted mortality using a simplified relationship between lapse and mortality:

$$\text{adjusted_qx} = \text{qx} \times (1 - \text{lapse_mortality_correlation} \times \text{lapse_rate})$$

This feature introduces interaction between lapse behaviour and mortality exposure. It should be interpreted as an educational simplification. In production actuarial work, mortality-lapse interaction would require careful empirical support and assumption governance.

10. Risk Adjustment Methodology

Under IFRS 17, the Risk Adjustment for non-financial risk represents the compensation an entity requires for bearing uncertainty about the amount and timing of cashflows arising from non-financial risk. In this project, the Risk Adjustment is implemented using a simplified stochastic confidence-level approach.

$$\text{RA} = \text{CoC ratio} \times \max(\text{quantile}(\text{PV liability}) - \text{mean}(\text{PV liability}), 0)$$

Risk Driver	Description
Mortality volatility	Uncertainty in death claims
Lapse volatility	Uncertainty in policyholder termination behaviour
Expense volatility	Uncertainty in future operating expenses
Counterparty default events	Uncertainty relating to reinsurer non-performance

This Risk Adjustment method is educational and simplified. It is not presented as a formally approved production method. A production IFRS 17 Risk Adjustment methodology would require board or committee-approved methodology, explicit confidence level or cost-of-capital framework, calibration to risk appetite, diversification methodology, sensitivity analysis, financial disclosure support, and independent actuarial review.

11. Contractual Service Margin Methodology

The model defines total liability as:

```
total_liability = BEL + RA
```

The initial Contractual Service Margin is calculated as:

```
initial_csm = max(-(BEL + RA), 0)
```

The onerous loss is calculated as:

```
onerous_loss = max(BEL + RA, 0)
```

An important modelling point is that BEL already includes premium inflows. Therefore, premiums should not be subtracted again in the CSM formula. Incorrect logic would double-count premiums. The model's CSM approach correctly uses the combined fulfilment cashflow position, BEL + RA, rather than applying a second premium adjustment outside BEL.

The model uses simplified coverage units:

```
coverage_unit(t) = in_force(t) x survival_ratio(t) x (1 - lapse_rate(t))
```

CSM release is allocated over the coverage period based on coverage units. This approach is suitable for a simplified term-life portfolio because coverage units approximate the pattern of insurance service provided over time.

12. IFRS 17 Grouping and Onerous Contract Testing

The grouping module assigns contracts into IFRS 17-style groups based on portfolio, cohort, profitability, risk, and onerous status. This demonstrates awareness that IFRS 17 measurement is not only a policy-level calculation exercise; contracts must also be grouped for recognition, measurement, and reporting.

Grouping Dimension	Description
Portfolio group	Business portfolio or product grouping
Annual cohort	Issue year or annual cohort classification
Profitability bucket	Onerous / profitable classification
Mortality risk group	Mortality risk classification
Lapse risk group	Lapse risk classification
Reinsurance risk group	Reinsurance exposure classification
Onerous status	Indicator of onerous contracts

Bucket	Meaning
Onerous	Contracts expected to produce a loss
No significant possibility of becoming onerous	Profitable contracts with lower onerous risk
Other profitable	Profitable contracts not classified in the previous category

```
onerous_status = 'onerous' if BEL + RA > 0
```

13. Scenario Analysis

Scenario analysis is included to evaluate how valuation results change under shocked assumptions. This is valuable because insurance liabilities are sensitive to mortality, lapse, expense, and discount-rate assumptions.

```
Base assumptions
-> Apply scenario shock
```



```

-> Rebuild projection
-> Recalculate cashflows
-> Recalculate BEL
-> Recalculate RA
-> Recalculate CSM
-> Summarise scenario outputs

```

Scenario	Expected Directional Impact
Mortality increase	Higher expected claims and potentially higher BEL
Lapse increase	Lower future premiums and lower exposure; direction depends on product profitability
Expense increase	Higher expected outflows and higher BEL
Discount-rate decrease	Higher present value of future liability cashflows

Scenario Output	Description
Total BEL	Aggregate Best Estimate Liability under scenario
Total RA	Aggregate Risk Adjustment under scenario
CSM opening	Opening CSM under scenario
CSM closing	Closing CSM under scenario
Onerous count	Number of onerous policies or groups
Onerous loss	Aggregate onerous loss amount

14. Audit, Governance, and Reproducibility Features

Output File	Purpose
assumption_snapshot.json	Stores assumptions used in a model run
audit_report.json	Summarises run metadata and validation information
run_registry.csv	Records run-level registry information

- Run ID
- Creation timestamp
- Assumption hash
- Methodology version
- Approval status
- Locked-in discount rate disclosure
- Row counts
- Reconciliation totals
- Data lineage paths

These features help demonstrate good modelling discipline. They make it easier to answer which assumptions were used, when the model was run, which output files were created, whether row counts reconciled, whether key totals reconciled, whether the assumption set changed, and which methodology version was applied.

These are lightweight governance outputs, not a full production governance framework. A production IFRS 17 reporting environment would require formal model approval, user access controls, change request workflow, segregation of duties, review and sign-off documentation, source system data lineage, control evidence, audit committee reporting, and independent validation.

15. Validation and Testing Framework

The project includes an automated test suite using pytest. The test suite can be run using:

```
python -m pytest -q
```

Expected result:
17 passed

#	Test Area	Purpose
1	Module imports	Confirms core modules can be imported
2	Discount curve uniqueness	Checks uniqueness and integrity of discount curve keys
3	Projection schema	Validates expected projection columns
4	Discount curve merge integrity	Checks that projection and discount curves merge correctly
5	Cashflow and BEL smoke checks	Confirms basic cashflow and BEL calculations run
6	Float64 precision mode	Checks numerical precision handling
7	PV cashflow component timing	Validates present value timing by cashflow component
8	Initial CSM formula	Checks CSM formula consistency
9	Grouping policy counts	Confirms policy counts are preserved through grouping
10	Opening survival ratio identity	Confirms Year 1 survival ratio equals 1
11	Premium and claim discount timing	Checks opening versus closing timing convention
12	BEL reconciliation	Reconciles BEL to adjusted liability cashflows
13	CSM release allocation	Validates CSM release allocation logic
14	Mortality shock direction	Confirms mortality stress moves results in expected direction
15	Audit snapshot determinism	Checks reproducibility of audit snapshots
16	Audit registry creation	Confirms run registry output is created
17	IFRS 17-style grouping fields	Checks presence of grouping classification fields

The automated tests provide evidence that the model has been checked for core mechanical consistency. They do not prove that the model is production-ready, but they demonstrate good coding discipline, awareness of model risk, reproducibility mindset, regression testing, actuarial control points, and the ability to validate valuation logic.

16. Key Strengths of the Project

The project separates major actuarial components into dedicated modules. This reflects how actuarial production models are often organised conceptually: assumptions, projection, cashflows, discounting, liability measurement, risk adjustment, CSM, grouping, scenarios, outputs, audit, and testing.

The project includes IFRS 17-related concepts such as fulfilment cashflows, BEL, Risk Adjustment, CSM, onerous testing, annual cohorts, profitability grouping, scenario analysis, audit snapshots, and locked-in rate disclosure. It does not claim full IFRS 17 compliance, but it demonstrates meaningful familiarity with the measurement structure.

Tool	Value
Python	General modelling language
pandas	DataFrame-based actuarial calculations
NumPy	Numerical operations
Pydantic	Configuration validation
pytest	Automated testing
openpyxl	Excel output support

The project is strongest when presented honestly: it is not production-ready, it uses synthetic data, it uses simplified IFRS 17 logic, it is an academic/CV portfolio project, and it demonstrates actuarial modelling potential.

17. Limitations and Future Improvements

Limitation	Explanation	Possible Mitigation
Synthetic policy data	Data does not come from a real insurer portfolio	Clearly disclose synthetic nature and focus on methodology
Assumptions not calibrated	Mortality, lapse, expense, and reinsurance assumptions are not based on insurer experience	Add sample experience study or calibration framework
Simplified mortality behaviour	Mortality is modelled using simplified rates and adjustments	Add mortality table selection, improvement, and underwriting classes
Simplified lapse behaviour	Lapse assumptions may not reflect real policyholder behaviour	Add duration-based, age-based, and premium-size lapse assumptions
Educational Risk Adjustment	RA is not formally approved or calibrated	Add confidence level documentation and calibration examples
Simplified CSM methodology	Does not fully model all IFRS 17 subsequent measurement complexities	Add locked-in rate accretion, changes in fulfilment cashflows, and loss component
IFRS 17-style grouping only	Grouping is simplified and not company-specific	Add formal portfolio definitions and annual cohort policy
No full disclosure package	Financial statement disclosures are not produced	Add IFRS 17 disclosure tables and movement analysis
Lightweight audit outputs	Audit files are metadata, not full governance workflow	Add review sign-off, approval status, and change logs
No independent peer review	Model has not been independently reviewed	Add review checklist and reviewer comments

Priority	Improvement	Rationale
1	Add detailed methodology documentation	Makes the project easier to review
2	Add GitHub Actions for automated tests	Demonstrates professional software workflow
3	Add more granular unit tests	Improves model reliability
4	Add assumption calibration examples	Strengthens actuarial credibility
5	Add mortality table documentation	Improves actuarial transparency
6	Add experience analysis module	Connects assumptions to observed experience
7	Add CSM movement analysis	Makes IFRS 17 reporting more complete
8	Add loss component tracking	Improves onerous group treatment
9	Add output dashboard	Improves interpretability of results
10	Add model validation report	Strengthens governance presentation

18. Conclusion

The IFRS 17 Term Life Valuation Engine is a strong actuarial science portfolio project. It demonstrates the ability to structure an actuarial valuation model in Python, apply simplified IFRS 17 measurement concepts, generate policy-level projections, calculate expected cashflows, discount liabilities, estimate BEL and Risk Adjustment, calculate CSM, perform onerous testing, create IFRS 17-style groups, run scenarios, produce audit outputs, and validate model logic through automated tests.

The project is appropriately positioned as an academic and CV project rather than a production IFRS 17 reporting system. This distinction is important and increases the credibility of the work. The use of synthetic data is also appropriate because real policy-level actuarial datasets are generally confidential and unavailable for public portfolio

projects.

From a recruiter, actuarial manager, or university instructor perspective, the project demonstrates practical actuarial modelling ability, Python programming capability, understanding of insurance cashflows, awareness of IFRS 17 measurement concepts, modular software structure, testing discipline, professional awareness of model limitations, and the ability to document actuarial methodology clearly.

Overall, the project is a credible and impressive actuarial science portfolio project for a 4th-year student, especially when presented honestly as a simplified educational IFRS 17 valuation engine.

Appendix A: Main Configuration Parameters

Parameter	Description
Number of policies	Size of synthetic portfolio
Projection years	Number of future years projected
Locked-in discount rate	Rate used for locked-in measurement concepts
Current discount rate	Rate used for current valuation
Premium amount	Policy premium assumption
Expense amount	Operating expense assumption
Acquisition cost	Initial acquisition cost assumption
Cost-of-capital ratio	Scaling parameter for simplified RA
Mortality assumption	Base mortality rate or table
Lapse assumption	Base lapse rate or curve
Mortality volatility	Stochastic mortality uncertainty driver
Lapse volatility	Stochastic lapse uncertainty driver
Expense volatility	Stochastic expense uncertainty driver
Reinsurance parameters	Ceding, recovery, and default assumptions
Scenario shocks	Mortality, lapse, expense, and discount-rate shocks

Appendix B: Output Files

Output	Format	Purpose
Projection output	CSV / Excel	Policy-year projection details
Cashflow output	CSV / Excel	Expected cashflows by policy and year
BEL output	CSV / Excel	Best Estimate Liability results
RA output	CSV / Excel	Risk Adjustment results
CSM output	CSV / Excel	Initial CSM and roll-forward results
Grouping output	CSV / Excel	IFRS 17-style group classifications
Scenario output	CSV / Excel	Scenario valuation results
assumption_snapshot.json	JSON	Assumption record for reproducibility
audit_report.json	JSON	Run-level audit and reconciliation metadata
run_registry.csv	CSV	Registry of model runs

Appendix C: Test Coverage Summary

The current automated validation suite includes 17 passing tests.

```
python -m pytest -q
```

```
17 passed
```

Test Category	Coverage
Imports	Confirms modules load correctly
Projection	Confirms projection schema and survival identity
Curves	Confirms discount curve uniqueness and merge integrity

Test Category	Coverage
Cashflows	Confirms cashflow generation and timing conventions
BEL	Confirms liability calculation and reconciliation
RA	Supports valuation integration checks
CSM	Confirms initial CSM formula and release allocation
Grouping	Confirms policy count preservation and grouping fields
Scenarios	Confirms mortality shock direction
Audit	Confirms snapshot determinism and registry creation

Appendix D: CV Project Description

Long CV / Portfolio Description

IFRS 17 Term Life Valuation Engine - Developed a modular Python-based actuarial valuation engine for term-life insurance, demonstrating simplified IFRS 17 measurement concepts including policy-level projection, mortality and lapse assumptions, cashflow generation, discounting, Best Estimate Liability, Risk Adjustment, CSM calculation and roll-forward, onerous contract testing, IFRS 17-style grouping, scenario analysis, audit snapshots, and automated validation testing. The project uses synthetic data to demonstrate actuarial modelling architecture while acknowledging that real policy-level insurance data is typically confidential.

Short CV Bullet Version

- Built a modular Python IFRS 17-style term-life valuation engine with policy-level projection, cashflows, BEL, Risk Adjustment, CSM, onerous testing, grouping, scenarios, audit outputs, and automated pytest validation.
- Implemented actuarial cashflow and discounting logic using pandas, NumPy, Pydantic, pytest, and openpyxl, with synthetic policy data suitable for an academic actuarial portfolio project.
- Added validation coverage for projection schema, discount timing, BEL reconciliation, CSM formula, grouping consistency, scenario directionality, and audit output determinism.